

Build a Rock Sorting Key

BUILD • Science • 40 minutes

I can apply a clear rock-grouping rule, check a mystery specimen and defend the placement with evidence.

Prepare before pupils arrive

Setup: Reuse the W9A coded set and prepare one visible criterion card, yes/no mats and Mystery T. Keep W10 scratch and water equipment out of sight until the next lesson.

Safety: Observation and card grouping only. Prevent curiosity from becoming an unsupervised scratch, drop or break test. Maintain a complete image-card route.

Equipment: Coded rock specimens or macro cards from W9A; yes/no mats; criterion cards for crystals, grains, bands/planes and holes; Mystery T; Sorting Key Case Card.

Safety: Keep pre-screened samples on trays. This is observation and card grouping only: no scratch, water, drop or break tests. A no-touch macro-card route is equivalent.

Fallback: Point or say/sign while an adult moves cards. The pupil owns the criterion and every placement decision.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Compare and group rocks by their appearance and simple properties.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Name two properties observed in W9A. Compare Code A and Code E using one feature.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	criterion — the feature that decides consistent — same meaning every time key — questions that sort Write one question that another person can answer by observing: 'Are visible bands or clear flat planes present?' Create YES and NO mats. Keep the question and its meaning visible between them.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Reveal Mystery T only after the reference set is placed. Apply the unchanged question and quote the mystery evidence.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Apply this fixed question to every card: Are visible bands or clear flat sheet-like planes present? Read the fixed criterion.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Create and operate one yes/no sorting key for the coded rock set, place Mystery T and then show how a second rule changes the groups. Record: A Sorting Key Case Card with one exact question, all reference placements, defended Mystery T placement and one complete regrouping rule.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Defend Mystery T's placement and explain what changed when the criterion changed. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: YES, because faint visible bands still meet the stated rule.

Yes. The rule says visible bands, not dark bands, so the same meaning places Code C in YES.

Misconception repair

Repaired. The same rocks can form different valid groups when a new observable criterion is stated and applied consistently.

Guided checkpoint

Yes. The rule says visible bands, not dark bands, so the same meaning places Code C in YES. Read the exact criterion. Do not add a new colour or strength condition halfway.

Code A

NO · NOT VISIBLE IN THIS VIEW. NO: the supplied view does not show the criterion feature. Do not sort by crystals when the fixed question asks about bands or planes.

Code B

NO · NOT VISIBLE IN THIS VIEW. NO: no bands or flat planes are visible in this supplied view. Use only the fixed criterion.

Code C

YES · BANDS / FLAT PLANES PRESENT. YES: faint visible bands still meet the unchanged criterion. The rule says visible bands; it does not say they must be dark.

Code D

NO · NOT VISIBLE IN THIS VIEW. NO: the feature is not visible in this view. A pale colour does not answer the bands-or-planes question.

Code E

YES · BANDS / FLAT PLANES PRESENT. YES: the specimen shows clear flat sheet-like planes. Read the second half of the criterion: bands OR clear flat planes.

Code F

NO · NOT VISIBLE IN THIS VIEW. NO: large holes are visible, but the fixed criterion feature is not. Do not change to a holes rule until every specimen has completed this key.

Mystery T

YES · BANDS / FLAT PLANES PRESENT. YES: Mystery T is placed using the same visible-band evidence as the references. Keep the boundary still when the mystery arrives.

Model limitation

One two-way appearance key only classifies specimens under its chosen rule. It cannot identify an unknown rock from one image or reveal hidden properties.

Independent evidence

A Sorting Key Case Card with one exact question, all reference placements, defended Mystery T placement and one complete regrouping rule.

Supported: Use one prepared yes/no question, place four reference cards and Mystery T, then say one evidence word for each. Keep the criterion and matching feature symbol visible. Offer one card at a time and exact adult scribing.

Standard: Write or choose one yes/no criterion, classify all references, place Mystery T and regroup with a second rule. Use: My criterion is __. Code __ goes in YES/NO because __.

Stretch: Evaluate two different keys for the same set and explain which is more useful for a stated display purpose. Judge clarity, consistency and what each criterion reveals or hides.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W9B_Rock_Sorting_Key_Do-1.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.